Proposed Solution

❖ Introduction

Under the theme CrisisNet: Rapid Disaster Response & Aid Coordination, this project proposes a centralized digital platform designed to improve emergency communication during disasters and road accidents.

The system focuses on reducing communication delays, eliminating language barriers, and ensuring that citizens receive clear, structured, and location-relevant alerts in real time.

❖ Problem Context

During disasters and highway accidents:

- Authorities release updates across multiple channels.
- Messaging is often inconsistent.
- Alerts are frequently issued only in English.
- Citizens struggle to understand what action to take.
- There is no mechanism to measure whether the message was understood.

This communication gap leads to confusion, panic, delayed response, and avoidable risk to human life.

❖ Objective of the Proposed Solution

The primary objective of the proposed solution is to:

- Ensure rapid dissemination of emergency alerts.
- Deliver structured, action-focused instructions.
- Provide multilingual accessibility.
- Display only region-specific alerts.
- Measure public understanding of emergency messages.

❖ Key Features of the Proposed System

1. Automatic Alert Generation

When a disaster or accident is identified, the system automatically creates a structured alert containing:

- Incident type
- Location
- Severity level
- Clear "Required Action" instructions

This reduces manual delay and ensures faster communication.

2. Location-Based Alert Display

Citizens can:

- Select their district, or
- Allow location detection

The platform displays only alerts relevant to that specific area.This prevents information overload and reduces panic caused by unrelated incidents.

3. Action-Focused Messaging

Instead of displaying long government notices, the system presents:

- Short
- Clear
- Action-oriented instructions

Each alert includes a structured "What to Do" section to guide citizens immediately.

4. Multilingual Support

To remove language barriers, alerts are provided in multiple languages such as:

- English
- Hindi
- Malayalam (extendable to regional languages)

This ensures inclusivity and better comprehension across diverse populations.

5. **Text-to-Speech Accessibility**

Each alert can be converted into audio form.

This feature supports:

- Elderly citizens
- Visually impaired individuals
- Low literacy populations
- Citizens under stress during emergencies

6. **Public Understanding Feedback**

After viewing an alert, users can mark:

- "Understood"
- "Not Understood"

This feedback helps authorities evaluate whether communication is effective.

7. **Administrative Monitoring Dashboard**

Authorities can monitor:

- Number of alerts issued
- Region-wise alert distribution
- Public understanding percentage
- Engagement trends

This supports improved decision-making and future disaster communication planning.

❖ Innovation and Uniqueness

The proposed solution differs from traditional alert systems because:

1. It centralizes communication into one structured platform.
2. It provides multilingual and audio accessibility.
3. It filters alerts based on user location.
4. It measures whether citizens actually understood the message.
5. It focuses on action clarity rather than information overload.

❖ Alignment with UN Sustainable Development Goals

SDG 11 – *Sustainable Cities and Communities*

The platform strengthens disaster resilience and improves emergency communication infrastructure.

SDG 13 – *Climate Action*

It supports rapid communication during climate-related disasters such as floods, cyclones, and heatwaves.

SDG 16 – *Peace, Justice and Strong Institutions*

It promotes transparent, structured, and accountable communication during crises.

❖ Expected Impact

- Faster emergency communication
- Reduced confusion during disasters
- Improved public understanding
- Inclusive access across languages
- Data-driven evaluation of communication effectiveness

❖ Conclusion

The proposed solution establishes a structured and inclusive disaster communication platform under the CrisisNet theme. By combining automated alert generation, multilingual support, location filtering, and public feedback mechanisms, the system enhances clarity, accessibility, and effectiveness of emergency communication.

This platform contributes toward building resilient communities and strengthening institutional response during crises.